

Analyze the scenario with the expanded rules and identify all potential deadlock scenarios. For each suggested problematic situation, prove that it can potentially create a deadlock by applying Coffman's conditions.

When you prove the possibility of deadlock, suggest a solution for how to implement it correctly or how to change rules in order to keep as much original functionality without creating deadlocks. If the solution doesn't exist in the realm of the given rules, explain exactly why.

1. The food in bowls is not infinite.
2. The old cat lady replenishes the food at regular intervals. If the bowl is occupied with a cat, she will wait for the cat to finish eating.
3. The cats are distinguished as females and males each with the different properties.
4. The bowls are now arranged in pairs.
5. Each cat, no matter whether it is male or female, never eats all the food from a bowl (the cat wouldn't eat some of the remaining food). There is a threshold for food that always remains after the cat's feeding.
6. The mouse can eat from a bowl if the other bowl in the pair is not occupied by a cat. If a mouse finishes the food in a bowl. It rests in the bowl.
7. Two female cats can eat together in the same pair of bowls.
8. If the female cat finishes eating, she can eat from the other bowl in the pair. If it is unoccupied, it waits for the other female cat to finish eating, while still occupying the bowl. If the other bowl doesn't have the food over the threshold, it leaves the bowl pair and waits another bowl with food over threshold.
9. The male cats can't stand each other; they never pick the bowl if the other one in the pair is occupied by another male cat. If they finish the feeding, they act like female cats.
10. If any cat gets hungry and it cannot pick any bowl with food over the threshold according to the rules above, it waits for the old cat lady to replenish the food. (The mice are not affected by cats waiting for food to be replenished.)
11. The old cat lady is afraid of mice; she won't replenish any bowl in a house if there is at least one mouse currently feeding.

Deadlock Scenario 1: Lady-Mouse Infinite Wait

Problem:

- Hungry cats wait for bowls to be replenished.
- Old lady refuses to replenish while mice are feeding/resting.
- Mouse keeps resting in bowl indefinitely.

Coffman Conditions:

- Mutual Exclusion: Lady and mouse cannot use bowl area simultaneously.

- Hold and Wait: Mouse holds bowl while cats wait for food.
- No Preemption: Lady cannot force mouse to leave.
- Circular Wait: Cat → Lady → Mouse → Cat.

Solution:

- Add mouse timeout or force mice to scatter when cats are waiting for replenishment.
- Breaks circular wait.

Deadlock Scenario 2: Female Cat Bowl Swap

Problem:

- Two female cats each occupy one bowl in a pair.
- Both finish and wait for the other bowl at the same time.
- Neither releases current bowl.

Coffman Conditions:

- Mutual Exclusion: One cat per bowl.
- Hold and Wait: Each cat holds one bowl while waiting for the other.
- No Preemption: Cats cannot force each other out.
- Circular Wait: Cat A → Bowl 2 → Cat B → Bowl 1 → Cat A.

Solution:

- Cats must release current bowl before waiting for another.
- Or enforce ordered bowl acquisition / try_lock behavior.

Deadlock Scenario 3: Male Cat Replenishment Standoff

Problem:

- Male cat occupies bowl pair and waits for adjacent bowl to be replenished.
- Lady waits for cat to leave before replenishing.
- Both wait forever.

Coffman Conditions:

- Mutual Exclusion: Cat occupancy blocks replenishment.
- Hold and Wait: Cat holds bowl while waiting for food.
- No Preemption: Lady cannot replenish occupied pair.
- Circular Wait: Cat → Lady replenishment → Cat leaves.

Solution:

- Allow lady to replenish unoccupied bowl in pair even if cat occupies neighboring bowl.
- Otherwise deadlock cannot be resolved under original rules.

Deadlock Scenario 4: Mouse-Cat Threshold Trap

Problem:

- Cat waits for food replenishment.
- Mouse occupies paired bowl.
- Lady refuses to replenish because mouse is present.
- Mouse does not leave because no cat is actively eating.

Coffman Conditions:

- Mutual Exclusion: Mouse blocks replenishment access.
- Hold and Wait: Cat waits for replenishment while mouse holds bowl.
- No Preemption: Mouse cannot be forcibly removed.
- Circular Wait: Lady → Mouse → Cat → Lady.

Solution:

- Expand scatter rule so mice flee when cats are waiting for that bowl pair, not only when cats are actively feeding.